Bihar Engineering University, Patna

B.Tech 1st Semester Exam-2022

Course: B.Tech. *Time: 03 Hours*

Code: 102102 *Subject: Mathematics-1(Calculus & Linear Algebra)* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

(ii) There are NINE questions in this paper.

(iii) Attempt FIVE questions in all.

(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven Question only): **[2 x 7 = 14]**

(a) Lagrange's mean value theorem can be proved for a function *f(x)* by applying Roll's mean value theorem to the function

(i) $\varphi(x) = f(x) + kx^2$ (ii) $\varphi(x) = f(x) - kx^2$

(iii) $\varphi(x) = f(x) + kx$ (iv) $\varphi(x) = \{f(x)\}^2 + kx^2$

(b) The function $f(x) = x(x+3)e^{-\frac{x}{2}}$ satisfy all conditions of Roll's mean value theorem in the interval [-3, 0]. Then the value of c is :

(i) 0. (ii) 1.

(iii) 2. (iv) - 2.

(c) If $\begin{bmatrix} 4/3 \\ 1 \end{bmatrix}$ is an eigenvector of $\begin{bmatrix} 2 & 4 \\ 3 & 1 \end{bmatrix}$. What is the associated eigenvalue?

(i) 4/3 (ii) 5. (iii) -2. (iv) None of the above

(d) If f is continuous on [-2.354, 2.354] then

(i) $\int_{-2.354}^{2.354} f(cosx)dx = 0$ (ii) $\int_{-2.354}^{2.354} f(cosx)dx = 2\int_{0}^{2.354} f(cosx)dx$

(iii) 2.354 (iv) None of the above

(e) Let $f(x) = |x|^{\frac{3}{2}},\ x \in R$ then

(i) f is uniformly continuous

(ii) f is continuous, but not differentiable at x = 0

(iii) f is differentiable and derivative of f is continuous

(iv) f is differentiable, but derivative of x is discontinuous at x=0

(f) If A(2) = 2i – j+2k, A(3) = 4i – 2j +3k, then $\int_2^3 A.\frac{dA}{dt}dt$ is

(i) 5 (ii) 10 (iii) 15 (iv) 20

(g) If $\nabla \times \vec{F} = 0$. Then it is called.

(i) Solenoidal (ii) Rotational (iii) Irrotational (iv) None

(h) If $3x + 2y + z = 0,\ x + 4y + z = 0,\ 2x + y + 4z = 0$ be a system of equations, then

(i) It is inconsistent

(ii) It has only the trivial solution x =0, y=0, z=0

(iii) Determinant of the matrix of coefficients is zero.

(iv) None of these

(i) If f(x) is continuous in the closed interval [a,b] and f'(x) exists in (a,b) and f(a) = f(b), then there exist at least one value c(a<c<b) Such that f'(c) = 0 is called

(i) Taylor's theorem (ii) Mac Laurin's theorem

(iii) Rolle's theorem (iv) Lagrange's mean value theorem

(j) The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ is

(i) 3 (ii) 2

(iii) 1 (iv) None of these

[7]

Q.2 (a) Show that:

$$\int_0^\infty x^2 e^{-x^4 dx} \times \int_\infty^\infty e^{-x^4} dx = \frac{\pi}{8\sqrt{2}}$$

[7]

(b) Use L' Hospital rule to find the following limits.

$$\lim_{x\to 0} \frac{\cos x - \ln(1+x) - 1 + xy^x}{\sin^2 x}.$$

Q.3 (a) Test the convergence of [7]

$$1 + \frac{3}{7}x + \frac{3.6}{7.10}x^2 + \frac{3.6.9}{7.10.13}x^3 + \frac{3.6.9.12}{7.10.13.16}x^4 + \cdots$$

(b) Examine the convergence of the series of which the general term is [7]

$$2^2 4^2 6^2 \dots \frac{(2n-2)^2}{3.4.5\dots(2n-1)2n} x^{2n}$$

Q.4 (a) Obtain the fourth-degree Taylor's polynomial approximation to $f(x) = e^{2x}$ *about* $x = 0$. Find the maximum error when $0 \le x \le 0.5$ [7]

(b) It is given the Rolle's theorem holds the function $f(x) = x^3 + bx^2 + cx, 1 \le x \le 2$ at the Point $x = 4/3$. Find the value of b and c. https://www.akubihar.com [7]

Q.5 (a) Evaluate $\int_0^\infty \log\left(x + \frac{1}{x}\right)\frac{dx}{1+x^2}$. [7]

(b) Discuss the convergence of the sequence whose n-th term is $a_n = \frac{(-1)^n}{n} + 1.$ [7]

Q.6 (a) If f(x) = log (1+x), x>0 using <Madaurin's theorem, show that for $0<\theta<1$, [7]

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3(1+\theta x)^3}$$

Deduce that $\log(1+x) = < x - \frac{x^2}{2} + \frac{x^3}{3}$ for x>0

(b) Using Taylor's theorem, express the polynomial $2x^3+7x^2+x-6$ in powers of (x-1). [7]

Q.7 (a) Find the values of a,b,c if $A = \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$ is orthogonal? [7]

(b) Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ and find its inverse. Also express $A^5-4A^4-7A^3+11A^2-A-10I$ as a linear polynomial in A. [7]

Q.8 (a) Expand long x in power of $(x-1)$ by Taylor's theorem. [7]

(b) By using Beta function evaluate $\int_0^1 x^5(1-x^3)^{10}dx.$ [7]

Q.9 (a) Find the Fourier series to represent the function defined as [7]

$$f(x) = \begin{cases} x + \frac{\pi}{2}, & -\pi < x < 0 \\ \frac{\pi}{2} - x, & 0 < x < \pi \end{cases}$$

(b) Evaluate: i) *div F* and *Curl F*, where $F = grad\,(x^3+y^3+z^3-3xyz)$. [7]

ii) If $F = (x+y+z)\,\hat{\imath} + \hat{\jmath} - (x+y)\hat{k}$ *show that F Curl F* $= 0$.

Bihar Engineering University, Patna

B.Tech. 1st Semester Examination, 2023

Course: B.Tech.
Code: 103102 *Subject: Mathematics-I(Calculus & Differential Equations)* *Time: 03 Hours*
Full Marks: 70

Instructions:-
(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.
(v) Symbols used (if any) have their usual meanings

Q.1 ***Choose the correct answer of the following (Any seven question only):*** [2 x 7 = 14]

(a) The solution of the given differential equation is :
$(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$
(i) $x3 + y3 - 6xy(x + y) = c$ (ii) $2x^3 + 4y^3 - 4xy = c$
(iii) $x^3 + y^3 - 5xy(x + y) = c$ (iv) $x^2 + y^2 - 6xy = c$

(b) The complementary function for the given differential equation is:
$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = x^2$$
(i) $c_1e^{-x} + c_2e^{-4x}$ (ii) $c_1e^{-2x} + c_2e^{-x}$
(iii) $c_1e^{-3x} + c_2e^{-x}$ (iv) None of the above.

(c) The solution of the differential equation $\frac{dy}{dx} - y\cot(x) = 2x\sin(x)$ *is* :
(i) $y = x^2 + c$ (ii) $y \sin(x) = 2x + c$
(iii) $y \operatorname{cosec}(x) = x^2 + c$ (iv) $y \cot(x) = 3x + c$

(d) The wronskian of the functions {sin(x), cos(x) } is :
(i) 1 (ii) -1
(iii) 0 (iv) 2

(e) Suppose $\vec{A} = x^2z^2\,\hat{\imath} - 2y^2z^2\,\hat{\jmath} + xy^2z\,\hat{k}$ then the divergence of $\vec{A}$ at point P (1, -1, 1) is :
(i) 4 (ii) 5
(iii) 6 (iv) 7

(f) Let $A=\int\int_R \sin(x + y)\,dx\,dy$, where the region R is given by $\{0 \le x \le \frac{\pi}{2}, 0 \le y \le \frac{\pi}{2}\}$ then the value of A is :
(i) -1 (ii) 1
(iii) 0 (iv) 2

(g) The series $1 + \frac{3}{2!} + \frac{5}{3!} + \frac{7}{4!} + \dots\dots\dots\dots\dots\dots$ is:
(i) convergent (ii) divergent
(iii) Neither convergent nor divergent (iv) None

(h) If f:R → R be a function defined by $f(x) = \begin{cases} x^2 + \sin\frac{1}{x^2}, & x \neq 0 \\ 0, & x = 0 \end{cases}$
(i) f is differentiable on R (ii) f is differentiable on R – {0}
(iii) f is not differentiable at 0 (iv) None of these

(i) If $f(x, y) = e^{x+y+1}$ then the value of $\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$ is
(i) $(x + y)\,e^{x+y+1}$ (ii) $xe^y + ye^x$
(iii) $2e^{x+y+1}$ (iv) None of the above

(j) The value of $\Gamma\left(-\frac{3}{2}\right)$ *is*
(i) 1 (ii) $\frac{\pi}{2}$
(iii) $\sqrt{\pi}$ (iv) $\frac{4}{3}\sqrt{\pi}$

Q.2 (a) Show that: [7]

$$\int_0^1 [\ln x]^n dx = (-1)^n n!,\ n \epsilon N.$$

(b) If α and β are the roots of the equation $ax^2+bx+c=0$ the evaluate the following limit. [7]

$$lim_{x\to\alpha} \frac{1-\cos(ax^2+bx+c)}{(x-\alpha)^2}$$

Q.3 (a) Test the convergence of [8]

$$\frac{1}{2\sqrt{1}} + \frac{1}{3\sqrt{2}}x^2 + \frac{1}{4\sqrt{3}}x^4 + \frac{1}{5\sqrt{4}}x^6 + \ldots\ldots\ldots\ldots$$

(b) Prove that [6]

$$Curl(f\boldsymbol{v}) = (grad\ f) \times \boldsymbol{v} + f\ curl\ \boldsymbol{v}$$

Q.4 (a) Use Taylor's theorem to prove that [8]

$$1+\frac{x}{2}-\frac{x^3}{8} < \sqrt{1+x} < 1+\frac{x}{2}, if\ x>0.$$

(b) If p(x) is a polynomial of degree > 1 and k ∈ R, prove that between any two real roots of p(x) = 0 there exists a real root of $p'(x)+kp(x) = 0$. [6]

Q.5 (a) Show that the function f(x,y) = $\begin{cases} \frac{x^3+y^3}{x-y}, x \neq y \\ 0, \quad x = y \end{cases}$, is not continuous at origin. But f_x and f_y exists at (0, 0). 4 [6]

(b) Show that $\int_0^{\pi/2} sin^m\theta cos^n\theta d\theta = \frac{\Gamma(\frac{m+1}{2})\Gamma(\frac{n+1}{2})}{2\Gamma(\frac{m+n+2}{2})}$. 2 [8]

Q.6 (a) Find the maxima and minima of the function $u = x^3+3xy^2-15x^2-15y^2+72x$. 5 [7]

(b) Find $u(x,y) = cos^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right), 0<x,y<1$, then prove that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+\frac{1}{2}$ cot u = 0. [7]

Q.7 (a) Find the Fourier series to represent the function defined as 4 [6]

$$f(x) = \begin{cases} x+\frac{\pi}{2}, -\pi<x<0 \\ \frac{\pi}{2}-x, 0<x<\pi \end{cases}$$

(b) Solve the following differential equations, [4x2=8]

(i) $(x+2)^2\frac{d^2y}{dx^2}-(x+2)\frac{dy}{dx}-3y=0$

(ii) $(y-1)dx + x(x+1)dy = 0$

Q.8 (a) Use variation of parameters method to solve [7]

$$(D^3+1)y = e^{x/2}\sin\frac{x\sqrt{3}}{2}$$

(b) Using operator method solve the following differential equation, [7]

$(D^2-2D+5)\ y = e^x \sin\ 2x$ https://www.akubihar.com

Q.9 (a) Find a complete integral of the following equation [7]

$$p^2x+q^2y=z$$

(b) Find the general solution of the partial differential equation [7]

$$x^2\frac{\partial z}{\partial x}+y^2\frac{\partial z}{\partial y} = (x+y)z$$

Bihar Engineering University, Patna

B.Tech. 1st Semester Examination, 2023

Course: B.Tech. | *Time: 03 Hours*
Code: 105102 | *Subject: Mathematics-I (Calculus and Linear Algebra)* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin
(ii) There are NINE questions in this paper
(iii) Attempt FIVE questions in all
(iv) Question No. 1 is compulsory.
(v) Symbols used (if any) have their usual meanings

Q.1 *Choose the correct answer of the following (Any seven question only):* [2 x 7 = 14]

(a) The value of $\int_0^1 x^3 (1-x^2)^{5/2} dx$ is

(i) $\frac{\pi}{2}$ (ii) $\frac{2}{63}$ (iii) 1 (iv) None of the above

(b) The area $\iint_R \sin(x+y)\, dxdy$ over R $\{(x,y); 0 \le x \le {}^{\pi}/_{2}, 0 \le x \le {}^{\pi}/_{2}\}$

(i) 0 (ii) 1 (iii) 2 (iv) -1

(c) The matrix $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ is

(i) Invertible but not orthogonal (ii) Invertible and orthogonal
(iii) Neither invertible nor orthogonal (iv) None of the above

(d) The sequence $\{a_n\}$, where $a_n = \sqrt[n]{n}$ is

(i) Is not convergent.
(ii) Convergent and $\lim_{n \to \infty} a_n = 0$.
(iii) Convergent and $\lim_{n \to \infty} a_n = 1$
(iv) None of the above

(e) Let $f(x) = \cos(x^2), X \in R$ then

(i) f is uniformly continuous
(ii) f is continuous, but not uniformly continuous
(iii) f continuous but unbounded
(iv) f is Lipschitz continuous

(f) $\lim_{(x,y) \to (1,2)} \frac{x^2+7y}{x+y^2}$

(i) Does not exists
(ii) Exists and the value is 2
(iii) Exists and the value is 3
(iv) None of the above

(g) If $A = 2\hat{\imath} + \alpha\hat{\jmath} + \hat{k},\ B = \hat{\imath} + 3\hat{\jmath} - 8\hat{k}$. Then A and B are orthogonal if

(i) $\alpha = -2$ (ii) $\alpha = 2$ (iii) $\alpha = -1$ (iv) $\alpha = 1$

(h) Let $x + 2y + z = 0, 3x + 4y + z = 0, x - z = 0$ be a system of equations. Then

(i) It is inconsistent
(ii) It has only the trivial solution $x = 0, y = 0, z = 0$
(iii) Determinant of the matrix of coefficient is zero
(iv) None of these

(i) The value at which the Rolle's theorem is applicable for $f(x) = \cos\frac{x}{2}$ in $[\pi, 3\pi]$ is

(i) $\frac{5\pi}{2}$ (ii) $\frac{3\pi}{2}$ (iii) 2π (iv) none of the above

(j) The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$ is

(i) 3 (ii) 2 (iii) 1 (iv) none of these

Q.2 (a) Show that : [7]

$$\int_0^{\frac{\pi}{2}} \log(\tan x + \cot x)\, dx = \pi \log 2$$

(b) Find the following limit. [7]

$$\lim_{n \to \infty} \left[t \ln\left(1 + \frac{3}{t}\right) \right]$$

Q.3 (a) Test the convergence of [8]

$$x^2 + \frac{2^2}{3.4} x^4 + \frac{2^2 4^2}{3.4.5.6.7.8} x^8 + \cdots \ldots$$

(b) Examine the convergence of the series of which the general term is [6]

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

Q.4 (a) Prove that $\log(1+x) = \frac{x}{1+\theta x}$, where $0 < \theta < 1$. Hence deduce. [8]

$\frac{x}{1+x} < \log(1+x) < x$.

(b) It is given the Rolle's theorem holds the function [6]

$f(x) = x^3 + bx^2 + cx, 1 \le x \le 2$

At the point $x = 4/3$. Find the value of b and c.

Q.5 (a) Evaluate $\int_0^{\pi/2} \frac{1}{a\cos^2 x + b\sin^2 x} dx$ [7]

(b) Discuss the convergence of the sequence whose *n-th* term is $a_n = \frac{(-1)^n}{n} + 1$. [7]

Q.6 (a) Show that the equation of the evolute of the curve [7]

$x^{2/3} + y^{2/3} = a^{2/3} is\ (x+y)^{2/3} + (x-y)^{2/3} = 2a^{2/3}$.

(b) Find the areas of the regions that lies inside the circle $r = a\cos\theta$ and outside the cardioid $r = a(1 - \cos\theta)$. [7]

Q.7 (a) Find the solution of the following system of equations [8]

$$x_1 + 2x_2 + 2x_3 = 2$$
$$x_1 + 8x_3 + 5x_4 = -6$$
$$x_1 + x_2 + 5x_3 + 5x_4 = 3$$

Also find the basis and dimension of the solution space.

(b) Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & 4 \\ 2 & 5 \end{bmatrix}$ and find its inverse. [6]

Also express $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$ as a linear polynomial in A.

Q.8 (a) Expand $\sin x$ in Taylor's series about $x = \frac{\pi}{2}$. [6]

(b) By using Beta function evaluate $\int_0^1 x^4 (1 - \sqrt{x})^5 dx$. [8]

Q.9 (a) Develop the Fourier series on $-\pi < x < \pi$ to represent the function defined as [6]

$$f(x) = \begin{cases} 0, -\pi < x < 0 \\ \pi, 0 < x < \pi \end{cases}$$

(b) (i) Evaluate div $\vec{F}$ and $Curl\ \vec{F}$, where [8]

$\vec{F} = grad\ (x^2 + y^2 + z^2) e^{-\sqrt{x^2+y^2+z^2}}$.

(ii) Let $\vec{F} = (-4x - 3y + az)\hat{i} + (bx + 3y + 5z)\hat{j} + (4x + cy + 3z)\hat{k}$. Find the values of a, b and c such that F is irrotational.

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Bihar Engineering University, Patna

B. Tech. I[st] Semester Examination, 2024

Course: B. Tech.
Code: 100102

Subject: Engineering Mathematics –I

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option the following (Any seven question only): **[2 x 7 = 14]**

(a) Which operation is not an elementary row operation?

(i)	Swapping two rows	(iii)	Multiplying a row by zero
(ii)	Adding a multiple of one row to another	(iv)	Multiplying a row by a non-zero constant

(b) The rank of a matrix is:

(i)	Number of non-zero rows in echelon form	(iii)	Number of rows
(ii)	Number of columns	(iv)	None of the above

(c) The inverse of a matrix using Gauss-Jordan method involves:

(i)	Only determinant calculation	(iii)	Reducing matrix to diagonal form
(ii)	Augmenting with identity matrix and applying row operations	(iv)	None of the above

(d) A matrix is diagonalizable if:

(i)	It has repeated eigenvalues only	(iii)	It has a full set of linearly independent eigenvectors
(ii)	It is symmetric	(iv)	It is square

(e) A matrix is similar to a diagonal matrix if:

(i)	It has complex entries	(iii)	It is invertible
(ii)	It has n linearly independent eigenvectors	(iv)	None of the above

(f) A function is Riemann integrable if:

(i)	It is continuous	(iii)	It is discontinuous everywhere
(ii)	It has infinite discontinuities	(iv)	None of the above

(g) Jacobian is:

(i)	A scalar	(iii)	A vector
(ii)	A matrix of partial derivatives	(iv)	A function

(h) Taylor's expansion for multivariable function includes:

(i)	Only first order terms	(iii)	Only second order
(ii)	All orders	(iv)	Only constant term

(i) Change of variables in double integral involves:

(i) Matrix algebra	(iii) Jacobian
(ii) Laplace transforms	(iv) Only constant term

(j) Triple integral over a region gives:

(i) Area	(iii) Volume
(ii) Length	(iv) Angle

Q.2 (a) What is the difference between a vector space and a subspace? Illustrate with examples. [7]

(b) Define and explain rank, row space, and column space of a matrix with examples. [7]

Q.3 (a) Explain the Cayley-Hamilton Theorem and verify it for a 2x2 matrix. [7]

(b) Define and compute the Jacobian for transformation $x = u + v$, $y = u - v$. [7]

Q.4 (a) Explain the Gauss-Jordan method for finding the inverse of a matrix with an example. [7]

(b) Define Hermitian, Skew-Hermitian, and Unitary matrices. Provide examples for each. [7]

Q.5 (a) Explain and prove the Rank-Nullity theorem. [7]

(b) Describe the method of finding eigenvalues and eigenvectors of a matrix. [7]

Q.6 (a) State and prove Rolle's Theorem with a graphical example. [7]

(b) Define Beta and Gamma functions. Derive the relation between them. [7]

Q.7 (a) Derive the Taylor series for a function of two variables. [7]

(b) Solve a maxima-minima problem using second derivative test. [7]

Q.8 (a) Evaluate a double integral to find area under a curve. [7]

(b) Convert a double integral into polar coordinates and evaluate. [7]

Q.9 **Write Short notes on any two of the following:** **[7 x 2 = 14]**

(a) Properties of Eigen vectors

(b) Riemann Integration & Riemann Sum

(c) Application to area and volume using double and triple integral

(d) Scalar and vector fields